

BOILER STEAM CONSUMPTION & CHEMICAL EFFICIENCY

FIELD AUDIT / MEASUREMENT INSTRUCTIONS

J Lieb Foods - Forest Grove, Oregon
Prepared from the Aqua Analytics boiler service report dated August 12, 2026

PRIMARY QUESTION

How much steam does the factory actually produce/consume, and how much boiler-treatment chemical is being used relative to that steam load?

IMPORTANT: The existing water meter gives makeup-water consumption. Makeup water alone does **not** equal total steam production when condensate is returned and water is lost through blowdown.

1. WHAT WE ALREADY KNOW

Item	Report value	Field significance
Water meter	6,253,811.7 gal current 6,165,009 gal previous	88,802.7 gal increase since 7/23/2026
Reported daily average	4,440.14 gal/day	Makeup-water baseline; NOT confirmed steam production
Feedwater temperature	150 F	Report states feedwater needs to be 180 F
Feedwater conductivity	1,335 umhos	Report calls high FW conductivity abnormal
Boiler conductivity	4,547 umhos	Report target shown: 2,500-3,500
Boiler chloride	100 ppm as Cl	Report target shown: 30-65
Boiler sulfite	36	Report target shown: 60-80
Boiler condition	Overcycled / insufficient blowdown	Report recommends increasing blowdowns
Softener / feedwater	Salt leaving softener	Report recommends increasing softener rinse time
C41 treatment pump	Changed 12 sec -> 6 sec	Record this setting before making further dosing comparisons

Source note: Values above are transcribed from the Aqua Analytics report dated August 12, 2026. The report also notes foaming and a strong chemical odor in the feedwater and says overall chemical results were nominal while the feedwater conductivity was abnormal. Do not treat the 4,440.14 gal/day makeup-water average as measured steam flow.

2. WHAT MUST BE MEASURED TO ANSWER THE STEAM QUESTION

Required measurement	How to obtain it	Why it matters
Makeup water	Record boiler water-meter start/end readings at the same time each day	Establishes new water entering the boiler system.
Boiler runtime	Record burner/boiler operating hours for the same measurement period	Converts daily totals into operating-hour rates.
Condensate return	Use a condensate-return meter if installed. If none exists, measure/estimate return of condensate by a deaerated and converted to steam repeatedly	Returned condensate can be deaerated and converted to steam repeatedly
Blowdown	Record continuous blowdown flow if metered. For manual blowdown, log blowdown duration and volume with the steam flow	Blowdown events, duration and volume with the steam flow
Feedwater	Record feedwater temperature at consistent operating conditions.	Needed to track system condition and report's 180 F recommendation.
Chemical use	Record tank/drum level or weight at start/end plus pump settings.	Allows chemical consumption to be normalized to water and steam load.
Boiler load	Record steam pressure and, if available, burner firing rate or steam-flow meter	Provides operating context; a steam-flow meter is the preferred direct measurement

3. FIELD PROCEDURE - RUN THIS AUDIT FOR AT LEAST 7 NORMAL PRODUCTION DAYS

1. Pick one consistent daily reading time. Use the same time every day.
2. Before changing chemical settings, document the current chemical products, pump names, stroke/speed/timer settings, drum levels, and any controller setpoints.
3. Record the makeup-water meter reading.
4. Record boiler operating hours for the previous 24 hours.
5. Record feedwater temperature, boiler pressure, feedwater conductivity, boiler conductivity, and other treatment tests normally used by the site.
6. Record every manual blowdown event. If there is continuous blowdown, determine its actual flow rate if safely measurable.
7. Record condensate return quantity if a meter exists. If no meter exists, mark it UNKNOWN rather than inventing a percentage.
8. Record chemical drum/tank usage. A marked sight level, scale/weight, calibrated tank volume, or verified pump delivery can be used.
9. Note production shutdowns, sanitation/CIP, steam leaks, failed traps, unusual venting, boiler trips, water-softener regeneration, or major load changes.
10. After the audit period, calculate averages. Compare chemical usage against BOTH makeup water and estimated/measured steam.

4. DAILY FIELD LOG

Date / Time	Water Meter (gal)	Boiler Run Hrs	FW Temp (F)	Boiler PSI	FW Cond.	Boiler Cond.	Condensate Return	Blowdown Qty	Chemical Used

Daily notes / unusual conditions:

5. BLOWDOWN & CONDENSATE AUDIT

Do not guess these values. If a measurement cannot be obtained, write UNKNOWN and identify what meter/test is needed.

Manual / intermittent blowdown log

Date/Time	Valve / Location	Duration	Flow if known	Calculated gallons	Reason / Notes

Condensate return check

Question	Answer / Measurement
Is there a condensate-return flow meter?	
Meter/tag/location	
24-hour condensate return, gal/day	
Condensate tank level/volume method used	
Known steam users that do NOT return condensate	
Known leaking/failed steam traps or open vents	

6. CALCULATIONS

A. Makeup water used (gal) = Ending water-meter reading - Starting water-meter reading

B. Makeup water rate (gal/day) = Makeup gallons / number of days

C. Makeup water mass (lb/day) = Makeup gal/day x 8.34 lb/gal (approximate water density for field estimating)

D. Blowdown gallons = Blowdown flow (gal/min) x total blowdown minutes

E. Condensate return % - calculate only when total condensate returned and total steam/condensate basis are actually measured on a consistent basis. Do not back-calculate a 'precise' percentage from makeup water alone.

F. Steam production - BEST METHOD: use a calibrated steam-flow meter and totalize lb/day or lb/hr. If no steam meter exists, steam must be estimated from a defensible mass balance using measured makeup, condensate return, blowdown, and other

known losses. Document all assumptions.

G. Chemical per 1,000 gal makeup = Chemical used / Makeup gal x 1,000

H. Chemical per 1,000 lb steam = Chemical used / Steam lb x 1,000

BASELINE REFERENCE ONLY - NOT A STEAM-FLOW RESULT

Reported makeup-water average = 4,440.14 gal/day. Approximate makeup-water mass = 4,440.14 x 8.34 = **37,031 lb/day**. This is the mass of NEW water entering the system, not proof that steam production is 37,031 lb/day. Condensate reuse and blowdown must be accounted for.

7. CHEMICAL DOSING AUDIT

Chemical / Product	Pump / Tag	Current Setting	Start Qty	End Qty	Used	Basis / Notes
C41		6 sec (per 8/12 report)				
GWT-16						
Other						
Other						

Normalize each chemical two ways:

1. Chemical per 1,000 gallons of makeup water - useful for treatment-feed comparison.
2. Chemical per 1,000 lb of measured/defensibly estimated steam - useful for comparing chemical consumption against actual plant steam load.

Do not change treatment solely from this worksheet. Use the site's approved water-treatment limits, treatment provider guidance, boiler manufacturer requirements, and actual water-test results. This audit is intended to establish consumption and dosing baselines.

8. CURRENT RED FLAGS TO VERIFY

- Feedwater temperature was 150 F; report states 180 F is needed.
- Boiler conductivity was 4,547 umhos against the report's 2,500-3,500 range.
- Feedwater conductivity was 1,335 umhos and was specifically called abnormal.
- Report states boiler is overcycled and insufficient blowdown is causing sludge deposits/carry-over; recommendation was to increase blowdowns.
- Report observed foaming and an overwhelming chemical smell in feedwater.
- Report says salt/chloride is leaving the softener and recommends increasing softener rinse time.
- C41 pump was reduced from 12 seconds to 6 seconds on August 12; verify present setting before evaluating chemical consumption.
- Treatment pumps were found unprimed during the service visit and were reprimed; verify reliable chemical delivery before drawing dosing conclusions.

9. FINAL ANSWER SHEET

Audit period	_____ to _____
Average makeup water	_____ gal/day
Average boiler runtime	_____ hr/day
Measured condensate return	_____ gal/day OR UNKNOWN
Measured/estimated blowdown	_____ gal/day OR UNKNOWN

Measured steam total (preferred)	_____ lb/day; meter/tag: _____
Estimated steam total (if required)	_____ lb/day; method attached: YES / NO
Average steam rate while operating	_____ lb/hr
Condensate return percentage	_____ % OR UNKNOWN
Chemical #1 normalized use	_____ per 1,000 gal makeup / _____ per 1,000 lb steam
Chemical #2 normalized use	_____ per 1,000 gal makeup / _____ per 1,000 lb steam
Major losses/problems found	_____
Recommended follow-up	_____

Source: Aqua Analytics, J Lieb Foods boiler report, August 12, 2026, pages 1, 3, and 4. This field audit preserves the report's stated observations and targets. Any steam-production value not obtained from a steam-flow meter must be labeled as an estimate and accompanied by the measurement method and assumptions.